

EGAP Latin American Regional Hub. Learning Days: Introducción al diseño de investigación

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- ① Introduction
- ② Causal model
- ③ Inquiry
- ④ Data Strategy
 - Panel survey experiment
 - Instruments
- ⑤ Answer strategy
- ⑥ Results
- ⑦ Discussion

What are we going to learn?

- Research Design: four steps.
- To declare a research design.
- To read findings and to apply them.

MIDA framework (Blair et al. 2019)

- ① Causal Model of the world.
- ② Inquiry about the model.
- ③ Data strategy.
- ④ Answer Strategy.

Fuente: Blair, G., Cooper, J., Coppock, A., Humphreys, M. (2019). Declaring and diagnosing research designs. *American Political Science Review*, 113(3), 838-859.

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```

library(randomizr) # Para DeclareDesign
library(fabricatr) # Para DeclareDesign
library(estimatr) # Para DeclareDesign
library(DeclareDesign)
library(tidyverse)

# ----- Diseño aleatorio completo -----
# Diseño
# N=150
# Tamaño del efecto esperado: 0.15 SD.

# 1. Modelo
#####
# potential_outcomes: funcion de fabricatr
model <-
  declare_model(
    N = 1000,
    potential_outcomes(Y ~ runif(1, 0, 0.5) * Z + rnorm(N)))

```

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```
# 2. Pregunta
#####
inquiry <-
  declare_inquiry(ATE = mean(Y_Z_1) - mean(Y_Z_0))
```

Stylised fact

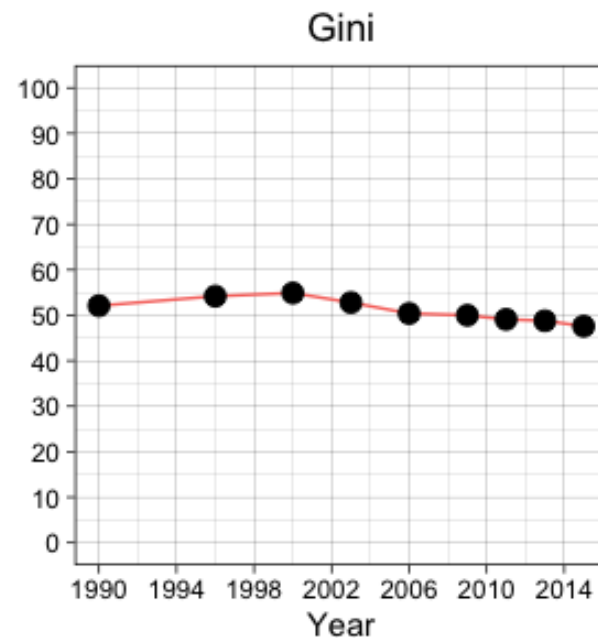
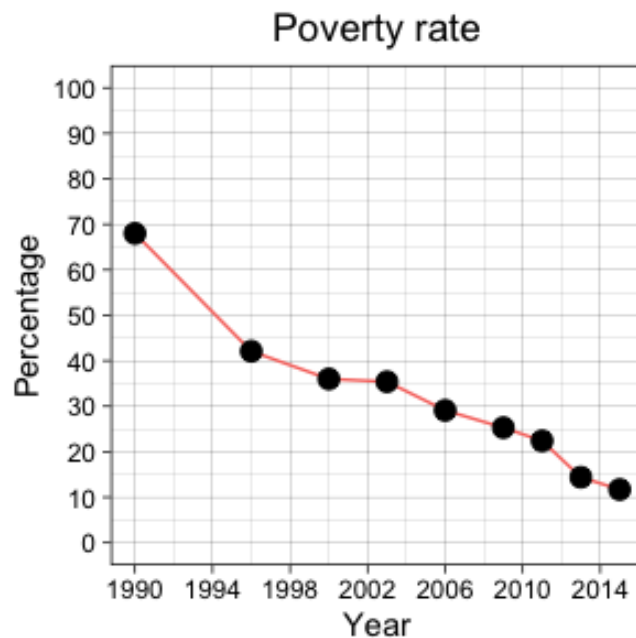
- Evolution of economic inequality in last decades.
- Citizens' knowledge about these trends.

What happens when researchers correct this misinformation?

- Contradictory findings:
 - Information about income inequality and social mobility heightens concerns about inequality and sometimes increases demand for redistribution (Alesina et al. 2018; Cruces et al. 2013; McCall et al. 2017; Karadja et al. 2017; Nair 2018).
 - Learning about income inequality affects perception of inequality but does not impact preferences for redistribution (Kuklinski et al. 2000; Kuziemko et al. 2015).
 - Information may make people more willing to accept large income inequality (Trump 2017).

- Reasons for these results
 - ① **Conflation of several constructs:** 1) Redistributive preferences, 2) perceptions of economic inequality and justice, and 3) Opportunity beliefs (people's understanding of the causes of economic inequality).
 - ② **Learning:** new and accurate information about social mobility and income distribution should lead those with new information to correspondingly adjust their beliefs (Hauser and Norton 2017; McCall et al. 2017).
 - **Challenge 1:** Evidence for contexts with increasing economic inequality, but counterfactual?
 - **Challenge 2:** Social sampling. Awareness of, and exposure to, social structure is tightly "bounded by private orbits in which they live" (Mills 1959).
 - **Challenge 3:** Backlash. Relative to the prior belief, the posterior belief moves in the opposite direction of the information presented.

Chile as an adjudicative case



Source: PNUD 2017.

- **Our research:** In a context with decreasing poverty trends but persistent income inequality, how does information about economic inequality impact opportunity beliefs?
- Contributions.

Hypotheses: Learning

- H1a: Factual information about poverty reduction reinforces the meritocratic dream and therefore leads to believe more in individual than structural opportunity factors.

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```
# 3. Estrategia de los datos
#####
# reveal_outcomes: funcion de fabricatr
# complete_rs: funcion de randomizr
# complete_ra: funcion de randomizr

data_strategy <-
  declare_sampling(S = complete_rs(N = N, n = 150), filter = S == 1) +

  declare_assignment(Z = complete_ra(N)) +

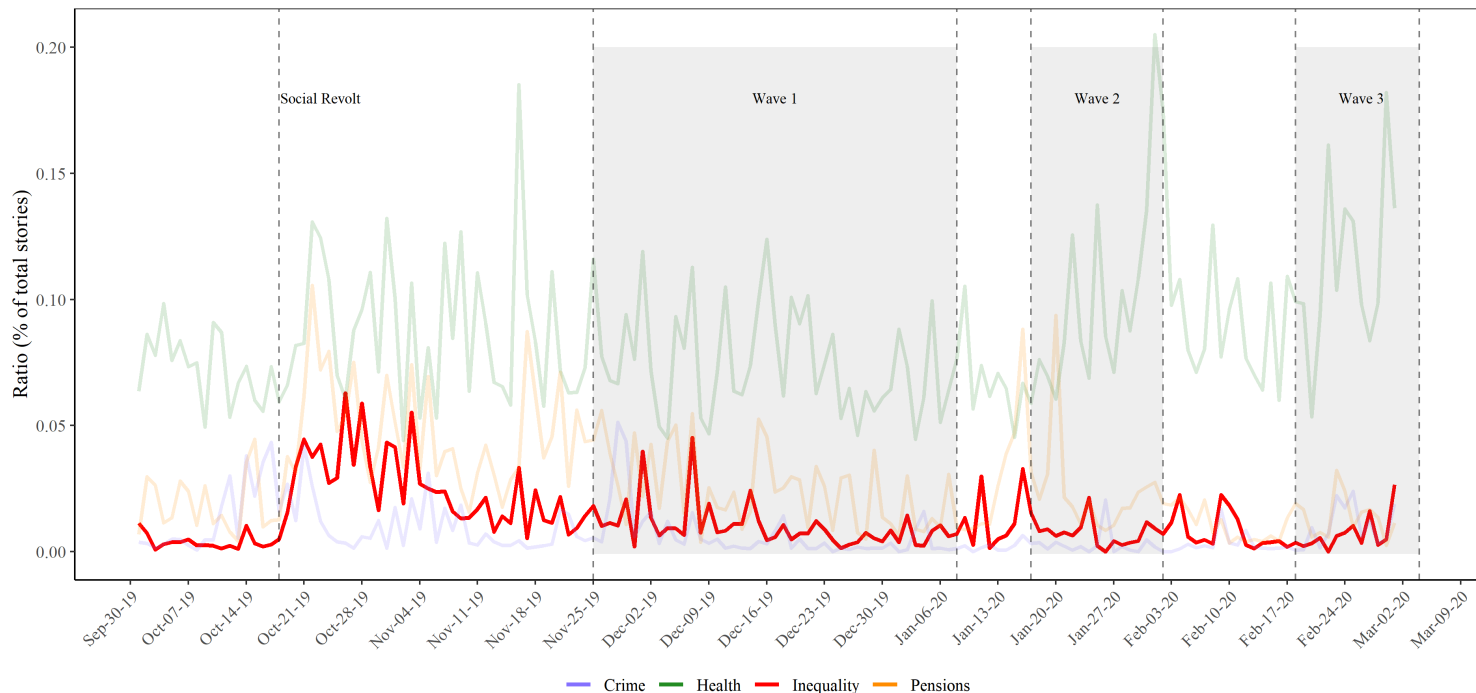
  declare_measurement(Y = reveal_outcomes(Y ~ Z))
```

TABLE 1. Comparisons of Experimental Designs

		T ₁		T ₂	
Post-only	R		X	O ₁	
	R			O ₂	
Pre-post	R	O ₁	X	O ₂	
	R	O ₃		O ₄	
Quasi-Pre-post	R	Q ₁	X	O ₁	
	R	Q ₂		O ₂	
Within	R	X	O ₁		O ₂
	R		O ₃	X	O ₄

Note: R = Randomization assignment to a group. O = observation of the dependent variable. Q = observation of a variable closely related to the dependent variable. X = exposure to a treatment. T = time of implementation.

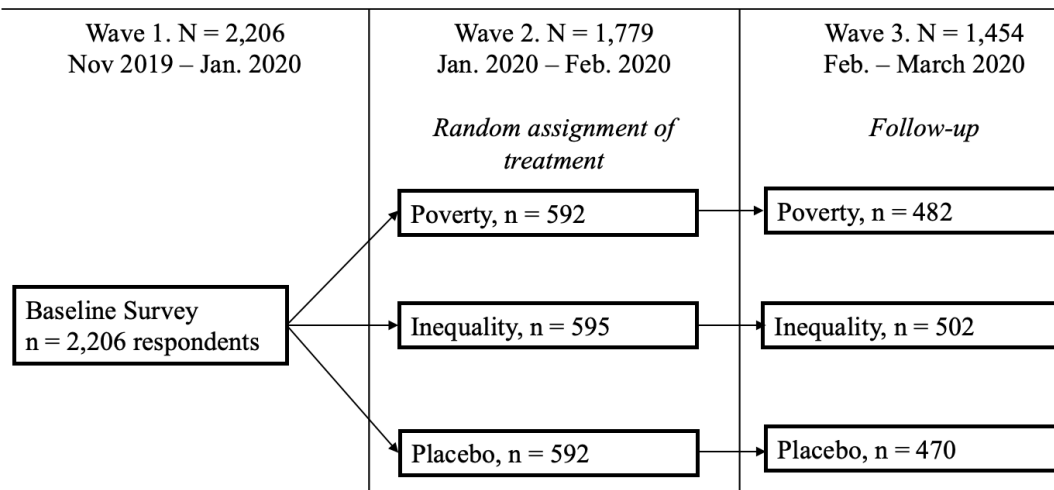
Fuente: Clifford, S., Sheagley, G., and Piston, S. (2021). Increasing precision without altering treatment effects: Repeated measures designs in survey experiments. *American Political Science Review*, 115(3), 1048-1065.



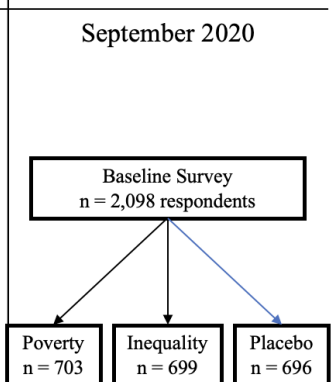
Source: Own elaboration based on Media Cloud data (<https://mediacloud.org/>).

Power: 0.80 for N=580 in each condition. Quotes for gender, age and education. N in wave 2 is 1779.

Study 1: Panel Survey. Nonprobability samples with quotas for age, gender and education



Study 2: Replication

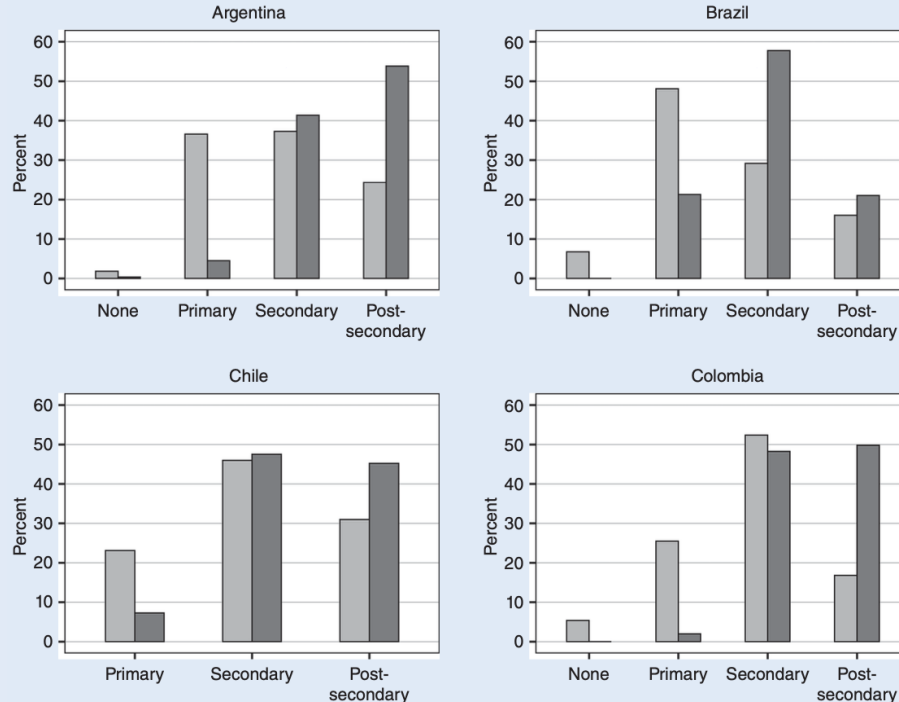


*Table 1***Sample Sizes and Fieldwork Dates**

COUNTRY (YEAR)	SAMPLE SIZE	FIELDWORK DATES
Argentina (2020)	1,207	March 23–April 27
Argentina (2021)	3,258	September 16–28
Brazil (2020)	1,250	March 23–April 30
Brazil (2021)	3,721	September 17–27
Chile (2021)	3,077	September 16–28
Colombia (2021)	3,238	September 16–28
Mexico (2020)	1,206	March 23–April 30
Mexico (2021)	3,252	September 16–27
Peru (2021)	3,019	September 16–28

Fuente: Castorena, O., Lupu, N., Schade, M., and Zechmeister, E. J. (2023). Online surveys in Latin America. *PS: Political Science and Politics*, 56(2), 273-280.

Comparing Population and Panel Education Distributions



Fuente: Castorena, O., Lupu, N., Schade, M., and Zechmeister, E. J. (2023). Online surveys in Latin America. *PS: Political Science and Politics*, 56(2), 273-280.

- Egami, N., and Hartman, E. (2023). Elements of external validity: Framework, design, and analysis. *American Political Science Review*, 117(3), 1070-1088.
- Porter, E., and Velez, Y. R. (2022). Placebo selection in survey experiments: An agnostic approach. *Political Analysis*, 30(4), 481-494.

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Treatments

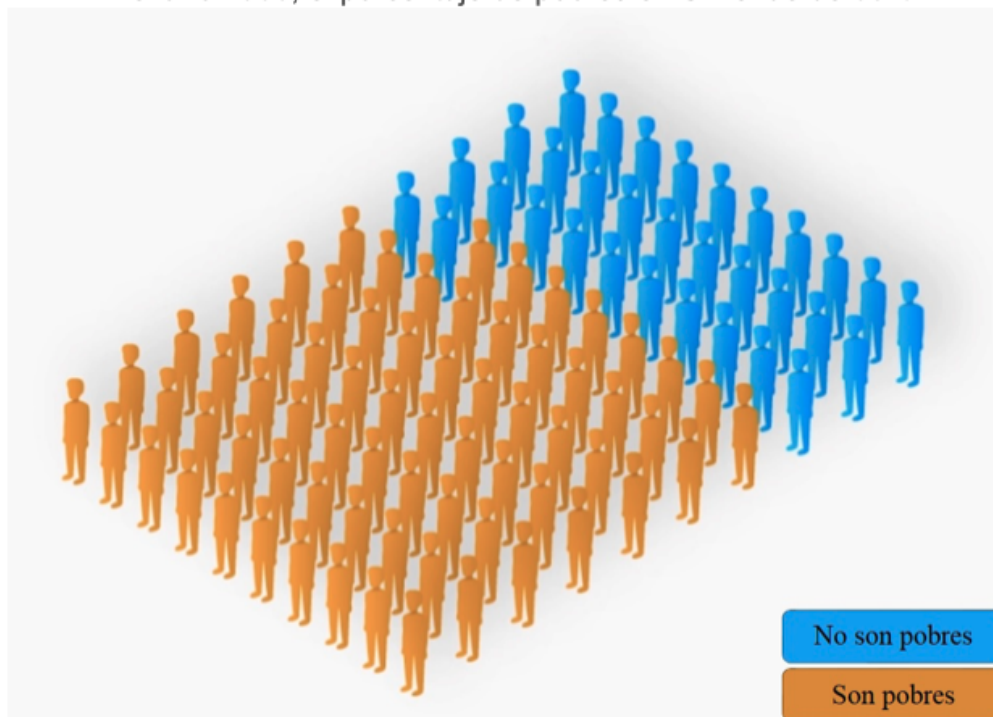
- Three conditions
 - ① **Poverty condition:** Information about evolution of poverty in Chile in period 1990-2015.
 - ② **Control condition:** Information about consume of cigarette in Chile during the last decades.
 - ③ **Inequality condition:** Information about income inequality in Chile in terms of index 10/10 that shows Chile as one of the countries with the highest income inequality within OECD.
- 80% of participants passes manipulation checks, 97% provides correct answer in attention check.

Poverty condition

Recent studies have shown **a decrease in poverty over the last decades in Chile**. According to a study by the United Nations Development Programme (2017), in parallel with accelerated and positive economic growth, **family income has increased, and access to goods has widened in Chile, which are signs of evident transformation of the living conditions**.

The following images summarize the main results of the study about poverty in Chile.

En el año **1990**, el porcentaje de pobres en Chile fue de **68%**



En otras palabras, de **100 personas**, **68 eran pobres** en 1990

Outcomes

- Opportunity beliefs: Battery of ISSP, social inequality module.
 - **Structural factors:** a) coming from a wealthy family.
 - **Individual factors:** b) hard work and c) composite index for education.

Covariates

- Socio-demographic characteristics, left/right scale, communes, and getting ahead measures for parents education and ambition.
- **Moderators:** egalitarianism, household income, and perception of poverty's evolution.
- All measures were applied before the treatments. Covariates are balanced among treatment conditions with the exception of gender and secondary.

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```

# 4. Estrategia de respuesta
#####
# difference_in_means: funcion de estimatr
# Ver: https://declaredesign.org/r/estimatr/articles/mathematical-notes.html

answer_strategy <-
  declare_estimator(Y ~ Z, .method = difference_in_means, inquiry = "ATE")

# Declaracion
#####
declaration_2.1 <- model + inquiry + data_strategy + answer_strategy

data<-draw_data(declaration_2.1)
data

```

```
> data
      ID      Y_Z_0      Y_Z_1 S Z      Y
1  0001 -1.5256812524  0.398367900 1 0 -1.525681252
2  0008  0.3611312537  1.953957484 1 1  1.953957484
3  0010 -0.1330065357  0.971710032 1 0 -0.133006536
```

```
# Notar la diferencia
```

```
mean(data$Y_Z_1) - mean(data$Y_Z_0)
```

```
summary(lm(Y~Z,data=data))
```

```
mean(data$Y[data$Z==1]) - mean(data$Y[data$Z==0])
```

$$Y_i = \alpha + \beta_1 T p_i + \beta_1 T i_i + \beta_2 X_i + \epsilon_i$$

- We estimate average treatment effects (ATEs) (β_1 and β_2) with OLS regressions (HC2 standard errors).
- We control for covariates X for efficiency. Furthermore, we use Benjamini-Hochberg correction for multiple comparisons.
- Causal heterogeneity:
 - Conditional average treatment effects (CATEs): OLS regressions with interactions.
 - We also explore heterogeneity by using random forests (Athey and Wager 2019; Brand et al. 2019; Molina and Garip 2019).
- Design and analysis are registered in pre-analysis plan in OSF.

```
> diagnose_design(declaration_2.1,sims=100)
```

Research design diagnosis based on 100 simulations. Diagnosis completed in 1 secs. 1

Design	Inquiry	Estimator	Outcome	Term	N	Sims	Mean	Estimand
declaration_2.1		ATE estimator	Y	Z		100		0.23
								(0.01)

Mean Estimate	Bias	SD	Estimate	RMSE	Power	Coverage
0.23	0.00		0.23	0.16	0.29	0.94
(0.02)	(0.02)		(0.02)	(0.01)	(0.04)	(0.02)

```
> program_diagnosands <-
+   declare_diagnosands(
+     success = mean(estimate > 0.10 & p.value < 0.05 & estimand > 0.10)
+   )
```

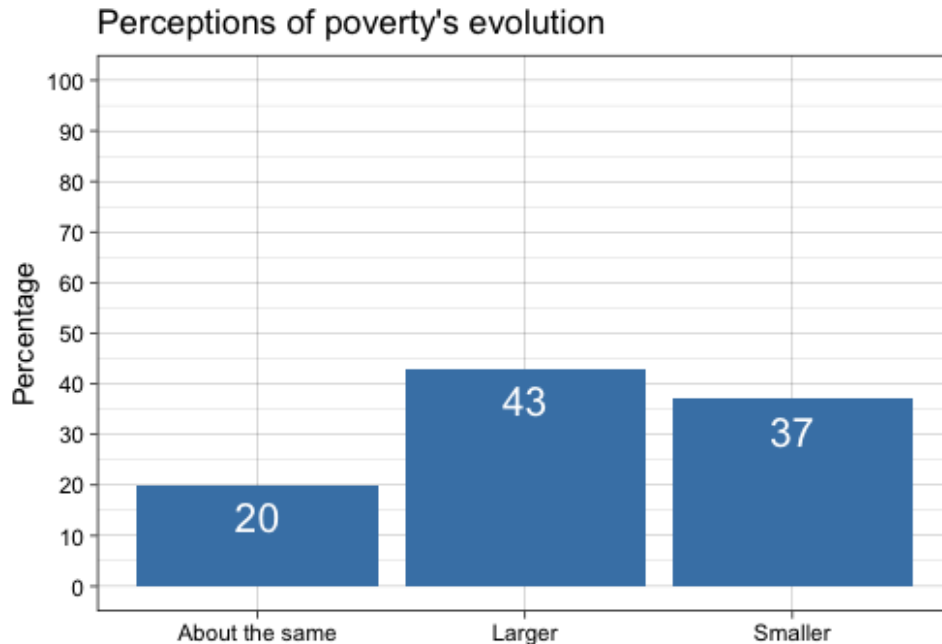
```
> declaration_2.1 |>
+ diagnose_design(diagnosands=program_diagnosands,sims=100)
```

Research design diagnosis based on 100 simulations. Diagnosis completed in 1 secs. 1

	Design	Inquiry	Estimator	Outcome	Term	N	Sims	Success
declaration_2.1			ATE estimator	Y	Z		100	0.37
								(0.05)

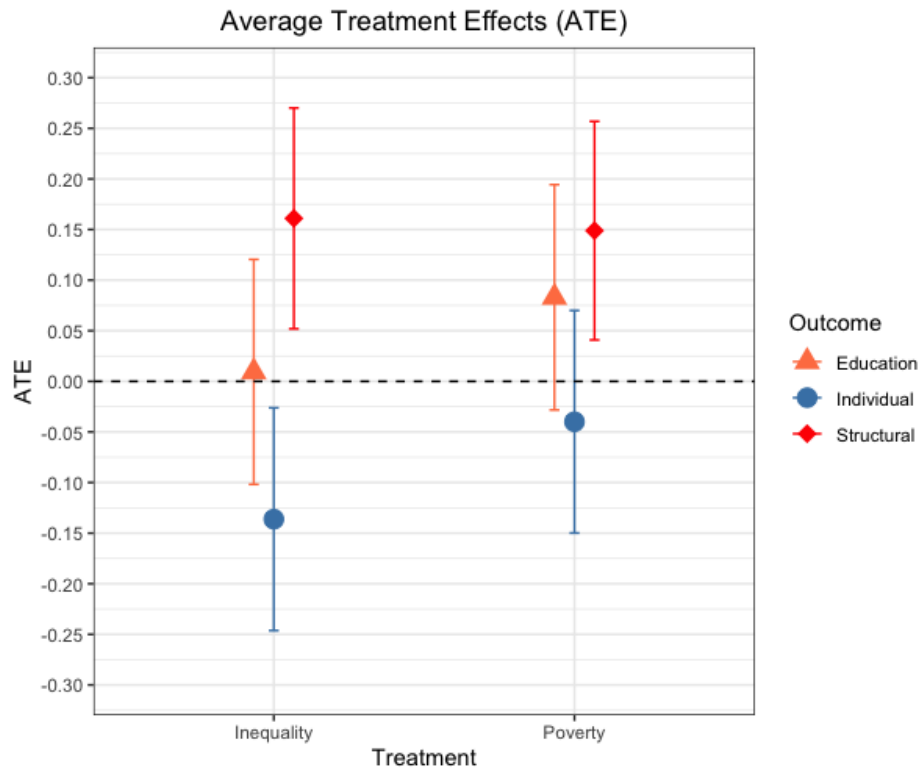
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Perception of economic inequality



Source: Fondecyt study (non representative of Chilean population). Question is: *Do you think that the quantity of poor people in Chile today is larger, smaller or about the same as it was 20 years ago?*

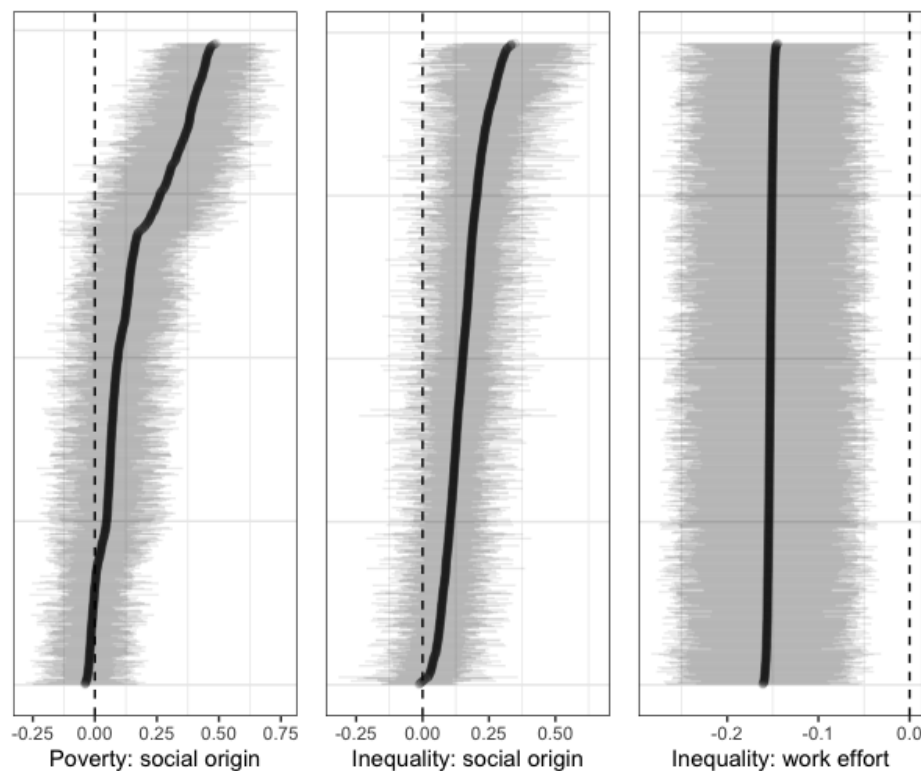
Learning



Note: Models include controls. Coefficients are standardized using Glass's delta.
N=1774.

Homophily/social sampling

Estimated effects from a forest estimator



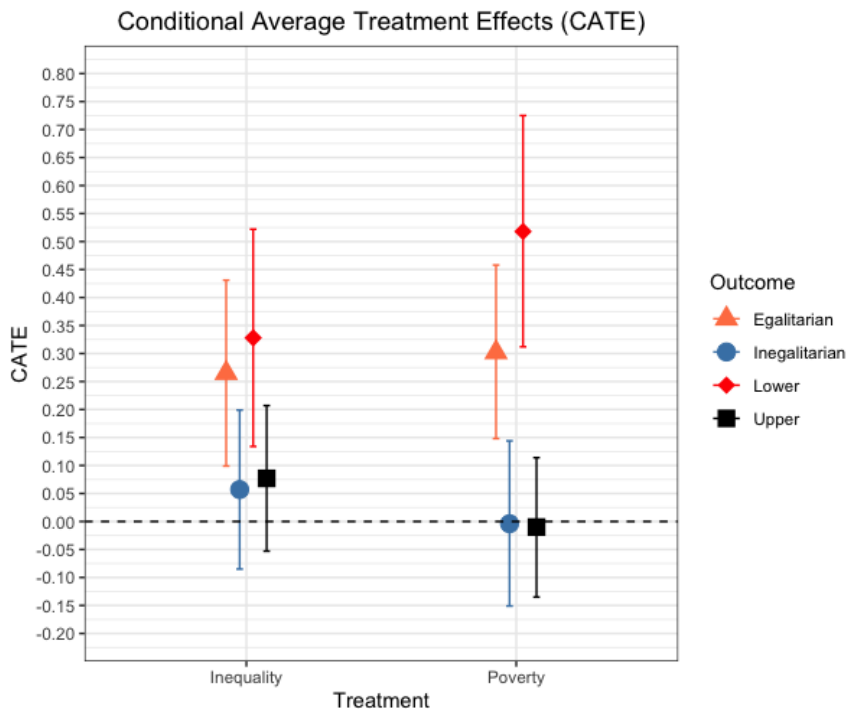
Homophily/social sampling

Table 1: Decomposition analysis of heterogeneity: Percentage of R squared

	Poverty: origin	Inequality: origin
Income	27.59	28.05
Egalitarianism	24.33	36.74
Poverty perception	7.69	5.76
Gender	17.21	2.77
Age	5.59	16.00
Education	14.25	8.16
Labor status	3.34	2.53
R squared	0.48	0.61
N	1180	1183
Shorrocks-Shapley decomposition		

Homophily/social sampling

Treatment effects by income groups and egalitarianism: **ONLY** belief in structural factor. N=1774.



Note: Models include controls. Coefficients are standardized using Glass's delta.

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- **Learning:** mixed evidence (H1b).
 - Factual information about large inequality increase support for structural factors and undermines the believe in meritocracy.
 - However, the poverty treatment also **increases** the support for the role of social origin in getting ahead in the life.
- **Homophily/social sampling:** evidence only for the lower end of the income distribution (H2a).
 - For low income groups, both treatments show very strong effects on the belief in structural factor.
 - Value attachment: egalitarianism.
- **Backlash:** The boomerang effect works for people at the lower end of the income distribution, but not only for individuals whose prior beliefs are opposed to the evidence presented.

- Future research: direct measure of economic homophily and external validity.
- Policy implications.